

Personalizing Activity Tracking: Enabling Older Adults to Teach and Train Their Wearables

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Abstract

Commercial activity trackers frequently fail to meet the needs of older adults by focusing on generic metrics (e.g., step counts) while overlooking the personally meaningful activities that they may want to track in their everyday life. This disconnect stems from a fundamental challenge: the difficulty of capturing accurate ground-truth labels for complex, real-world activities, which forces a reliance on generic, one-size-fits-all models. My doctoral research confronts this problem by developing a human-centered approach to recognize activities that truly matter to individual older adults, moving beyond basic posture detection to a rich, semantic understanding of daily life. Part of my completed doctoral work establishes the need for personalization by demonstrating significant variability in older adults' movement patterns and deconstructs the challenges of collecting data "in the wild," quantifying the costs and benefits of triangulating sensor data with user-provided verbal reports. This is complemented by technical explorations into data-driven methods for optimizing on-body sensor placement. Ongoing work investigates the feasibility of older adults training their own personalized models in-situ through transfer learning. My research will culminate in a framework for a "teachable" activity tracker that older adults can train themselves, and a neuro-symbolic model capable of inferring high-level, meaningful activities (e.g., "gardening," "playing golf") from a stream of low-level sensor data. Ultimately, this thesis contributes a new paradigm for activity recognition—one that shifts the focus from what technology can track to what people want to track, enabling the creation of personalized tools that genuinely support and reflect the lives of older adults.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing design and evaluation methods**; **Ubiquitous and mobile computing systems and tools**; **Empirical studies in ubiquitous and mobile computing**.

Keywords

Human Activity Recognition, Machine Learning, Older Adults, Accessibility, Wearables, Teachable Machine

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1 Problem Statement

Commercial activity trackers, while ubiquitous, have a low adoption rate among older adults [12, 14], perhaps due to a misalignment with their specific needs. Studies show that while many adults use trackers, adoption by those over 65 remains low [2, 11]. Indeed, in this thesis, we demonstrate that even basic activities such as walking require personalized cadence thresholds to be accurately classified, a feature absent in current systems. This points to a critical need for systems designed not with a one-size-fits-all approach, but through a human-centered process that prioritizes the specific needs, abilities, and motivations of older adults.

Activity tracking for older adults that goes beyond just counting steps has the potential to offer significant benefits [4]. Engagement in a diverse array of meaningful activities is associated with improved mental, physical, and social well-being [5, 18, 25]. However, by failing to recognize and validate these activities—from gardening to social outings—current technologies miss the opportunity to support what truly matters in the lives of older adults [25]. The motivation for this research is to engage older adult users in tracking their desired sedentary and non-sedentary activities by enabling them to teach their own human-centered activity tracker. In this way, they can reflect on and adjust their daily routines to better meet their health goals.

This doctoral research develops and evaluates a human-centered framework for teachable activity trackers that empowers older adults to train personalized models for recognizing the physical activities they deem meaningful.

- RQ1: What are the challenges and opportunities in collecting ground-truth activity data from older adults in real-world settings?
- RQ2: How can a teachable activity recognition system be effectively designed for older adults by leveraging co-design to align with their mental models of accuracy and control?
- RQ3: How feasible is it for older adults to successfully train a personalized activity recognition model in situ, and what strategies best support this process?
- RQ4: How can low-level sensor data be translated into high-level, semantically meaningful activity labels that reflect users' real-world experiences?



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2 Related Work

Activity tracking can be a valuable tool for supporting the health and well-being of older adults. While automatic tracking with wearable devices offers low effort, it often lacks the personal engagement and reflection fostered by manual self-tracking methods [15, 19]. While automatic tracking using wearables provide objective measurements, self-tracking captures subjective motivations and context [24].

Numerous studies in human activity recognition (HAR) have produced sophisticated models for posture detection [13, 17, 26]; however, dominant methodologies often treat the user as a passive subject rather than an active partner. In this paradigm, data is labeled by researchers, and used to train a static model, leaving no room for user input or personalization beyond the initial training phase [1, 21, 23]. Furthermore, the data used to train these HAR models is often collected in lab environments [1] that do not reflect the diversity of older adults’ real-world movement patterns. This entire approach overlooks the rich, contextual knowledge that only the user possesses.

This user-centered perspective also raises a question about the hardware itself. While using many sensors provides richer data [3, 22], it creates a trade-off between data fidelity and user burden. With wearables available in numerous form factors—from smartwatches and rings to thigh-worn sensors and smart shoes—it is unclear which body locations provide the most informative data for accurately capturing posture with minimal user burden. Prior work in sparse-IMU pose estimation, such as DIP-IMU [8] and IMU-Poser [16], has often relied on intuitive sensor placement. This highlights the need for a more evidence-based approach to answer a critical question: how can we build a robust yet unobtrusive data foundation for a truly human-centered tracking system?

My work challenges the dominant HAR paradigm by developing a “machine-teachable” approach, where the older adult transitions from being a data point to being the teacher with minimal effort.

3 Methods

Our proposed thesis is built upon two main approaches that explore the limitations of current activity trackers for older adults.

3.1 Approach 1: Shifting from Passive Subject to Active Teacher

The first principle is a fundamental shift in the user-system relationship: from the user as a passive consumer to an active director [9, 27]. This contrasts sharply with traditional HAR systems, which operate as immutable black boxes. Instead, we propose a “teachable” system—a concept informed by advances in areas like teachable object recognition [6, 7]—that operates as a collaborative partner. The mechanism is straightforward yet powerful: the older adult iteratively trains a personalized model by providing activity labels and, most importantly, correcting the system’s mistakes. This method directly addresses the desire for agency and control expressed by participants in our co-design studies, transforming the tracker from an opaque authority into a transparent and adaptable tool.

3.2 Approach 2: Beyond Postures: Tracking the Activities Older Adults Care About

The second principle is to move beyond tracking simple postures to recognizing what truly matters: the activities that older adults want to track. Commercial trackers ignore semantically rich activities like “gardening”, “shopping”, “cooking”, or “playing golf”, despite older adults expressing a clear interest in tracking them [10]. Recognizing these activities is challenging because they are not single postures, but rather composite sequences of atomic movements like sitting, standing, and walking. Our approach aim to addresses this by designing a system capable of collecting user-provided labels for these high-level activities and, ultimately, learning to distinguish them. This requires both a systematic effort in user-centered data collection and an advanced recognition system that can learn the unique temporal “fingerprint” of each composite activity.

3.3 The Proposed System Architecture

The proposed architecture, illustrated in Figure 1, is designed to operationalize these two principles. It is a human-in-the-loop system where data from a wearable device is processed by a personalized learning core. Through a user-driven labeling interface—designed based on our co-design findings—the older adult can teach the system, providing labels not only for basic postures but also for the high-level meaningful activities they wish to track. The system presents its predictions back to the user, who can then confirm or correct them. This interactive feedback loop provides a continuous stream of labeled data necessary to refine the personalized model, enabling it to learn both the user’s unique movement patterns and their personally significant activities over time.

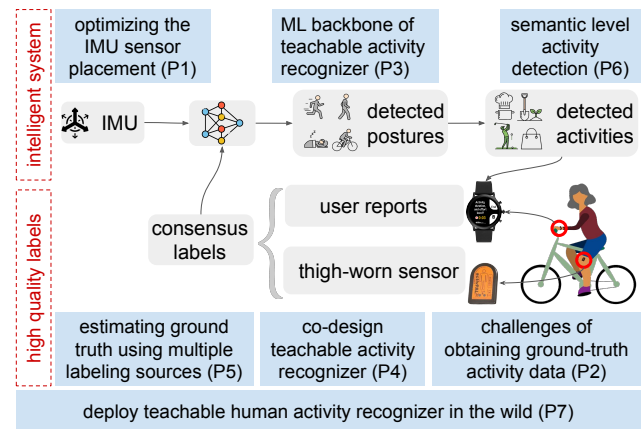


Figure 1: The proposed framework for a human-centered, teachable activity tracker. Each blue box corresponds to a specific research contribution, from unpacking ground truth challenges (P2) [10] and developing a core teachable machine learning platform (P4) to the final system deployment (P7).

4 Research Progress

My doctoral research is organized into three phases: completed, ongoing, and future work. This research comprises seven contributions (P1-P7), each corresponding to a published or planned

manuscript. My first-authored work is enriched and informed by collaborations on sensor optimization (P1) and co-design (P4).

4.1 Completed Work: Data-Driven Optimization of IMU Sensor Placement (P1)

To establish a robust data foundation, we developed a data-driven approach to identify optimal IMU sensor placements [20]. Our methodology involves training a high-performance Transformer model on a full set of 24 synthetically-generated IMU locations from motion capture data. We then employ model interpretability techniques, specifically feature ablation, to quantitatively assess which joints are most critical for accurate pose reconstruction. This process allows us to select a minimal, high-information set of sensor locations based on empirical evidence rather than intuition.

Our results show that this data-driven placement strategy significantly outperforms prior work. Critically, we also found that optimal locations are highly dependent on the type of activity being performed. This finding—that no single “best” configuration exists for all scenarios—reinforced our decision to prioritize ecological validity for our main user studies. Therefore, we selected two of the most common and socially acceptable sensor locations: the thigh, using the high-accuracy activPAL sensor, and the wrist, using a commercial smartwatch which also serves as the essential platform for user interaction and teaching.

4.2 Completed Work: Unpacking Ground Truth Challenges in the Wild (P2)

Our analysis of HAR studies from the past decade revealed a heavy reliance on researcher-led annotation, where activities are labeled either in controlled lab settings or by reviewing video recordings of participants [10]. This dominant paradigm raises two significant concerns: first, activities performed in a lab often fail to reflect the variability of real-world movements, and second, the use of video recording introduces substantial privacy issues. These limitations highlight the need for new, user-centered labeling methods that are both ecologically valid and privacy-preserving.

Building on our technical foundation and motivated by this methodological gap, our next study deployed a two-sensor configuration to critically examine the challenges of obtaining ground-truth labels for older adults’ activities in their natural environments [10]. A key finding was the clear need for personalization, even for seemingly simple activities. By analyzing step counts from the thigh-worn sensor, we found that cadence thresholds for distinguishing between low and moderate stepping intensity varied widely among participants, ranging from 80 to 105 steps/min. This demonstrates that a single, universal threshold for walking intensity is inadequate for this population and that models must be able to adapt to an individual’s unique movement patterns.

Furthermore, we analyzed the dual nature of using verbal reports as a labeling source. While they provide rich, semantic context that is missing from raw sensor data, they also introduce significant challenges, including imprecision in recalling time spans and a substantial manual annotation burden for researchers. To mitigate these issues, we developed a consensus labeling approach, triangulating the objective data from the high-accuracy thigh-worn sensor with the subjective, contextual data from verbal reports.

This triangulation, while powerful, comes with significant costs. We found that the process resulted in major data shrinkage, with only 34% of the collected smartwatch data being retained in the final consensus-labeled dataset. This loss was primarily due to a lack of verbal reports for low-intensity activities and disagreements between the two data sources, particularly for atypical postures (e.g., “sitting on a stool”). These findings highlight a critical challenge: achieving high-quality ground truth in the wild requires a trade-off between data quantity and data quality. This work, therefore, establishes the clear need for a new framework that can learn effectively from the limited, but high-quality, labeled data that can be realistically obtained in naturalistic settings.

4.3 Ongoing Work: Co-Designing with Older Adults to Inform System Design (P4)

To ensure our framework was grounded in user needs, I contributed to a co-design study, detailed in a manuscript currently under review, that explores older adults’ perceptions of providing high-quality labels in the context of training their own personalized activity trackers. Using the teachable machines paradigm as a scaffold, this study involved co-design sessions with 12 older adults to envision the process of labeling their activities, including describing activity names and time spans.

Our analysis revealed that older adults have a nuanced contextualized understanding of data quality that often differs from technical definitions. We unpack their unique perspectives on what constitutes acceptable accuracy, precision, and granularity for an activity tracker they would trust and use. For instance, participants valued semantic accuracy (e.g., correctly identifying “gardening”) over minute-by-minute postural precision. The study also surfaced participants’ preferred strategies for providing labels with minimal burden, highlighting a desire for a balance between user-initiated labeling for important events and intelligent, machine-initiated prompting at opportune moments. These findings directly motivated the core principles of our teachable framework, establishing the need for a system that not only empowers users with agency and a mechanism for correction but is also designed to bridge the gap between user perceptions and system requirements.

4.4 Ongoing Work: The Machine Learning Backbone: Training Strategies for Teachable HAR (P3)

My current research investigates the central hypothesis of this thesis: the feasibility of older adults training a personal activity tracker in-situ. This work explores the application of transfer learning techniques, where a generalized model is fine-tuned using a small number of labels provided by the user on their smartwatch. We are currently analyzing different training strategies and neural network architectures to determine the most effective and user-friendly methods for this process.

4.5 Future Work: Longitudinal Evaluation of a User-Trainable Activity Tracker (P7)

The next stage of this research will involve the development and longitudinal evaluation of the complete, teachable activity tracking application. A multi-week, in-the-wild deployment with older adults will assess how such a system can improve the automatic tracking of meaningful activities of the user's interest over time.

4.6 Future Work: Advanced Methods for Ground Truth and Semantic Interpretation (P5, P6)

Finally, two future technical explorations will address remaining challenges in semantic understanding. First (P5), we will develop a statistical model based on Hidden Markov Models to probabilistically fuse ambiguous labels from multiple sources, such as sensors and user reports. Second (P6), we will investigate a neuro-symbolic approach to infer semantically meaningful, composite activities from sequences of atomic postures, truly bridging the gap from posture to purpose.

5 Envisioned Contributions

This doctoral research aims to contribute key building blocks toward enabling older adults' agency in tracking activities they care about through human-centered AI on wearable devices.

- A rich empirical understanding of the challenges and opportunities in collecting ground-truth activity data with older adults in the wild.
- A personalized machine learning strategy for "teachable" activity recognition that demonstrate the feasibility of in-situ model training by older adults.
- A data-driven methodology for optimizing wearable sensor placement for pose estimation.
- A proof-of-concept "teachable" activity recognition system that re-frames the user's role from a passive subject to an active teacher.
- A model that detects high-level semantic activities from low-level sensor data.

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